

# MADHU PRIYA P

Electronics and Communication Engineering  
Student | Embedded & VLSI

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https://madhupriya1105.github.io/portfolio/



## SUMMARY

Versatile Electronics and Communication Engineering student with a strong foundation in core electronics, embedded systems, and IoT, along with programming and software development skills. Experienced in applying engineering concepts through projects and internships, with the ability to bridge hardware and software solutions. Adaptable, detail-oriented, and capable of working across multidisciplinary engineering environments.

## EDUCATION

### Dr NGP Institute of Technology, Coimbatore

BE | ECE

CGPA: 7.5

2023-2027

### shri vidhya bharathi matric hr. sec. school, Namakkal

Higher Secondary Education

Percentage: 83.3

2020 - 2023

## SKILLS

### Programming languages

C

C++

Embedded C

GitHub

Canva

Verilog (Basics)

Java

Circuit Simulation

### Embedded

Microcontroller Basics

PCB Design

Sensors & Actuators

### Tools

### Soft Skills

Problem Solving

Design Thinking

Team Work

### VLSI

Digital Electronics

Logic Design

## LANGUAGES

- Tamil - Native
- English - Proficient
- Hindi - Readable

## CERTIFICATIONS

- python - udemy
- AI Upskilling- Qualcomm
- VLSI Beginners-NIELIT
- Embedded Beginners-NIELIT

## EXPERIENCE

### Embedded Systems Intern

Robomiracle | 2025

- Worked on microcontroller-based embedded systems and basic robotics concepts.
- Interfaced sensors and actuators using Embedded C.
- Learned GPIO control, timing, and debugging of embedded applications.
- Assisted in testing and troubleshooting hardware modules.

## PROJECTS

### 1. Smart Parking System :

- Designed and implemented an automated parking system to manage vehicle entry and exit.
- Used RFID authentication for authorized access control.
- Integrated IR sensors for vehicle detection and slot availability.
- Controlled servo motor for gate automation using Embedded C logic.
- Displayed real-time parking status on LCD module.

Repository: <https://github.com/madhupriya1105/esp32-smart-parking-system>

### 2 Smart Ambulance Traffic Control System :

- Implemented RFID-based ambulance detection with IR sensor confirmation
- Designed traffic signal override logic using STM32 GPIO and timers
- Simulated RFID and sensor behavior in Wokwi for functional validation

## AREA OF INTEREST

- Embedded Systems
- VLSI Design
- Digital Electronics
- PCB Design